PURCHASE CERTAINTY ENGINE

Know the return before the checkout.

A multi-agent system that stress-tests fit, color, and fabric-to-skin safety on a digital mannequin of each shopper — and predicts, with a single Keep Probability score, whether a garment will be kept or returned.

LangGraph · 6 Agents | SSIM + CIELab Math | FastAPI + React | YouCam VTO + Skin

THE PROBLEM

Apparel e-commerce is bleeding money on returns.

THE COST

~30% return rates

Fashion returns run far higher than any other vertical — every return erases margin on a product that shipped, was worn, and was shipped back.

THE CAUSE

No fit confidence pre-purchase

Shoppers buy blind: the garment was never seen on their body, their skin, or in their colors. "Fit gamble" drives most returns.

THE BLIND SPOT

Merchants can't see it coming

Retailers learn about a bad SKU only after the returns arrive — no early-warning between catalog and cash register.

THE SOLUTION

One score. Three stress tests. A mannequin that knows you.

FIT

SSIM Fit Stress-Test

Three AI try-on renders of the garment on the shopper's actual photo; pixel-level SSIM variance exposes an unstable drape before it becomes a return.

COLOR

CIELab Color Harmony

Garment color matched against the shopper's seasonal palette and skin tone using Delta-E — kills the "it didn't suit me" returns.

FABRIC

Fabric-to-Skin Safety

Material map audited against personal allergies and skin concerns (rosacea, eczema). A flagged allergen multiplies the risk penalty.

The result is a **Keep Probability score (0–100)** with a plain-language verdict: STRONG_BUY · CONSIDER_CAUTION · HIGH_RETURN_RISK.

HOW IT WORKS

Six agents, running in parallel, converging on one verdict.

VTO Agent	Color Agent	Fabric Agent	Personalization	History Agent	Synthesis
3 AI renders → fit stability	ΔE vs seasonal palette	allergen / skin audit	fit bias · mood slider	past keep / return signals	weighted master score + LLM verdict

Keep Probability = **0.45 × fit** + **0.30 × color** + **0.25 × fabric**

+ personalization & purchase-history deltas × mood scalar

× **0.40** when a flagged allergen is detected

Every run persists as a session; shopper keep / purchase / return feedback closes the loop and retrains the engine's signals.

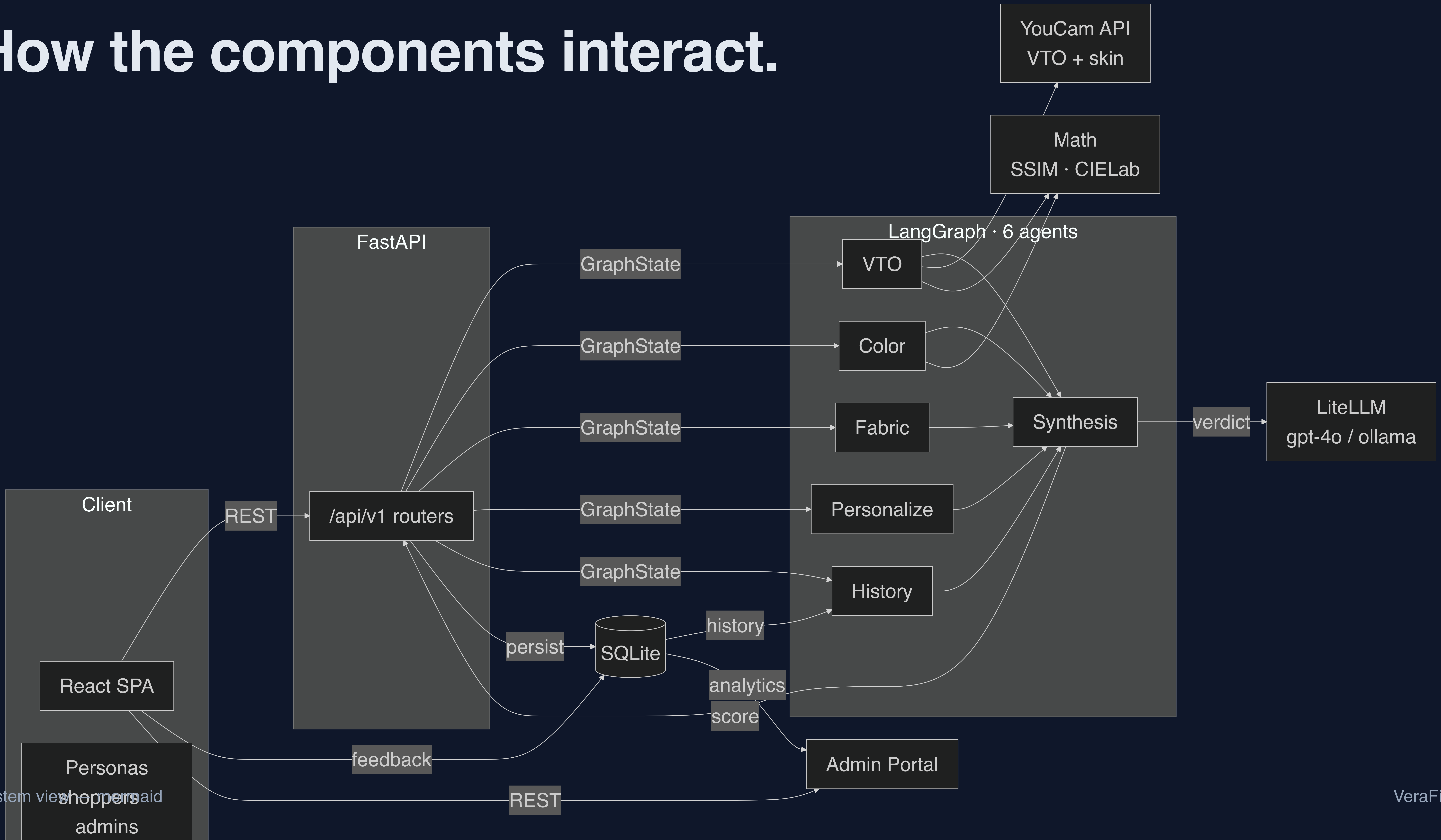
THE SIX AGENTS

Five experts run in parallel. One synthesiser decides.

#	AGENT	ROLE IN THE WORKFLOW	KEY OUTPUTS
1	**VTO Agent** vto_agent.py	Runs 3 parallel AI try-on renders of the garment on the shopper's actual photo (YouCam cloth-v4), stress-testing fit stability	renders · SSIM score · diff heatmap · unstable flag
2	**Color Agent** color_agent.py	Matches garment color to the shopper's seasonal palette & skin tone via perceptual CIELab ΔE	harmony score · ΔE · palette match · L*a*b*
3	**Fabric Agent** fabric_agent.py	Audits the material map against allergies & skin concerns (rosacea, eczema, friction, heat)	safety score · warnings · 0.40 penalty flag
4	**Personalization** personalization_agent.py	Applies fit preference, comfort-vs-style bias, and the mood slider	score delta · mood scalar (0.90–1.10)
5	**History Agent** history_agent.py	Aggregates past keep / return / purchase sessions into evidence-backed signals	learnings · recommendations · score delta
6	**Synthesis Agent** synthesis_agent.py	Fuses all five outputs into the weighted master score, picks the verdict, writes the AI explanation (LiteLLM)	keep_probability · verdict · explanation

SYSTEM ARCHITECTURE

How the components interact.



HYPER-PERSONALIZED FITMENT

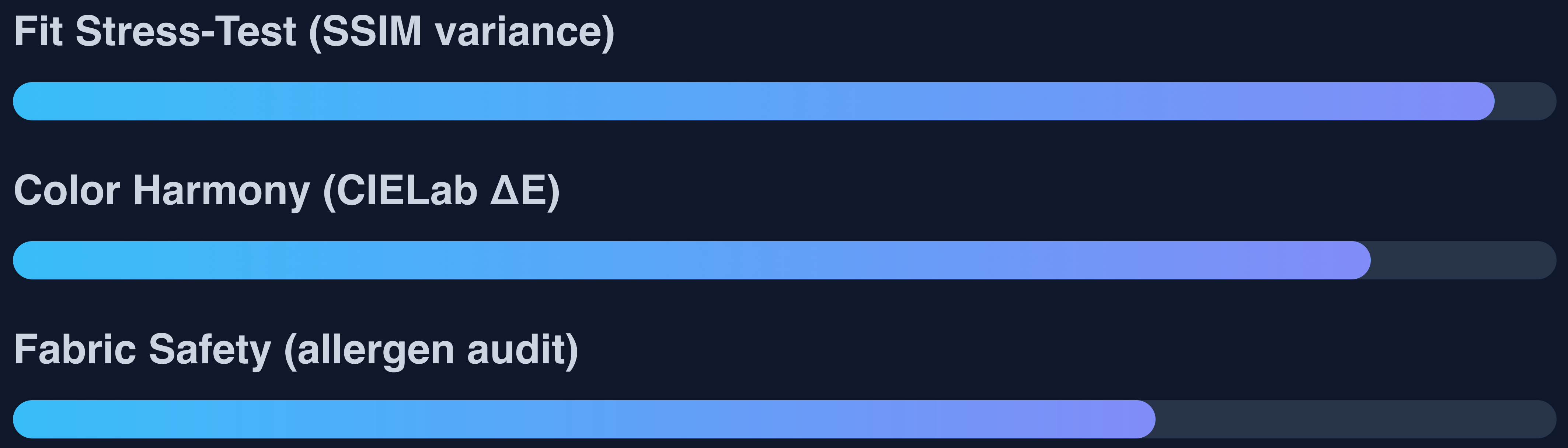
Real people, real photos, real predictions.

Each shopper has a calibrated digital mannequin — gender-appropriate, full-body, unique. The VTO drapes the garment onto **their** photo, not a generic model. Skin analysis runs on their selfie via YouCam.

- **Elena Vance** · luxury · Cool Winter · rosacea · wool + nickel
- **Astrid Holm** · eco-naturalist · Warm Autumn · eczema · synthetics / latex / wool
- **Lars Hedlund** · minimalist menswear · Warm Autumn · sensitive skin

Gender-aware

Men's garments get men's imagery; women's keep women's. Every mannequin differs — the prediction is per-person, per-SKU.



B2B MERCHANT PLATFORM

Two vendors. Two dashboards. Zero data bleed.

- Real-time fleet pulse & Keep Rate
  live per-vendor performance
- Vendor clusters + 7-day demand trends
  Venice vs Nordic fleets — strictly isolated
- Supplier defect analytics & AI-efficacy
  do predictions match reality per SKU?
- Inventory clearance audits + advice
  flag & clear high-return-risk stock early

Marcus Vance — Venice Luxury Atelier

Sensitive-skin executives · atelier eveningwear · cashmere

Freja Lindqvist — Nordic Organic Weaves

Eco-naturalists · cocoon comfort · wool-avoiders

A SKU-prefix resolver (SLK/CRP/CSH… → Venice · LNN/OVR/WOL… → Nordic) keeps every derived metric vendor-scoped end to end.

BENEFITS

Who wins when the prediction is right.

SHOPPERS

Buy with confidence

See the garment on your body, in your colors, against your skin before paying. Fewer wasted purchases — and an honest CONSIDER_CAUTION beats an expensive regret.

MERCHANTS

See returns before they happen

Early warning on high-return-risk SKUs, vendor-scoped fleet intelligence, and a measured keep-rate to prove the return-rate reduction.

THE INDUSTRY

Predict, don't react

A template for pre-purchase prediction with deterministic, auditable math — turning returns from a cost center into an explainable signal.

THE STACK

Built to ship — and to scale.

BACKEND

FastAPI · Python 3.14

Async SQLAlchemy + aiosqlite, pydantic-settings config, 30+ REST endpoints under /api/v1.

INTELLIGENCE

LangGraph · LiteLLM

Orchestrated multi-agent graph; verdicts via OpenAI / Anthropic / Gemini or local Ollama.

FRONTEND

React · Vite · TS

Fitting Room, Calendar, History, Learnings, Mannequin calibration, and the Merchant Admin.

RENDERING + SKIN

YouCam API

cloth-v4 virtual try-on and skin/facial-tone analysis, with deterministic mock fallback offline.

MATH

scikit-image · OpenCV

SSIM stability engine + CIELab color engine — deterministic, testable scoring kernels.

DELIVERY

Docker · AWS

Single-container image; one command to deploy on ECS Fargate + ALB or EC2.

ROADMAP

From certainty engine to returns-as-a-service.

SHIPPED · M1–M3

Core Engine → Analytics → Delivery

Multi-agent scoring, vendor-isolated B2B dashboard, container + AWS deploy, full documentation.

NEXT · M4

Hardening & Scale

Real auth + JWT, Alembic migrations, Postgres, S3 asset storage, HTTPS, CI/CD, background task queue.

M5

Deeper Prediction

On-device auto-calibration, generative 3D fit simulation, richer fabric knowledge base, price sensitivity.

M6

Network Effects

Anonymized fleet-intelligence marketplace, return-insurance scoring, supply-chain early warning, PWA rollout.

VERAFIT AI

Thank you.

Every returned garment is a failed prediction.
We make the prediction before the purchase.

Let's demo the Fitting Room

Ask about vendor analytics

Explore the architecture